

CNN-Based Water Pollution Image Classification for Environmental Monitoring: A Comparative and Honestly Reported Study of a Custom Convolutional Network and a MobileNetV2 Transfer-Learning Baseline on a Small Image Dataset

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Abstract

Water pollution is an escalating environmental problem that threatens public health and ecological stability, and it requires monitoring methods that can scale beyond slow, expensive laboratory testing. Conventional water-quality assessment depends on manual sampling and laboratory analysis, which limits how frequently and how widely contamination can be checked. This study investigates whether a lightweight convolutional neural network (CNN) can classify water images as clean or polluted, and whether a MobileNetV2 transfer-learning baseline generalizes better on the same small dataset. A custom CNN and a frozen-backbone MobileNetV2 model were trained under identical conditions on a 293-image dataset split into 234 training and 58 validation images using TensorFlow and Keras in Google Colab. The custom CNN reached 79.06% training accuracy but only 43.10% validation accuracy, while MobileNetV2 reached 50.00% validation accuracy, both close to the 50% chance level for a binary task. A detailed inspection of the data pipeline shows that the two class folders were produced by randomly splitting a single unlabelled image set, which means the reported labels do not encode a true clean-versus-polluted distinction and largely explains the near-chance performance. The principal contribution of this work is therefore methodological honesty: it documents a realistic small-data failure mode, traces it to

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a label-construction flaw, and shows that a pretrained model’s general visual features still transfer marginally better than scratch-trained features.

Keywords: Convolutional Neural Network, Water Pollution, Image Classification, Transfer Learning, MobileNetV2, Environmental Monitoring, Explainable AI

Highlights

- A custom CNN and a MobileNetV2 baseline are compared on a small 293-image water dataset.
- The custom CNN overfits (79.06% train versus 43.10% validation accuracy).
- MobileNetV2 with a frozen backbone reaches 50.00% validation accuracy.
- A label-construction flaw in the data pipeline is identified and reported transparently.
- Findings show the bottleneck is data quality and labelling, not model architecture.

1. Introduction

1.1. Application Background and Practical Importance

Rivers, lakes, reservoirs, and coastal waters across the world face mounting pressure from industrial discharge, plastic waste, agricultural runoff, and untreated sewage. Contaminated water is not only an ecological problem but also a direct public-health risk wherever communities rely on it for drinking, irrigation, or fishing. The conventional response to suspected contamination is to collect a physical sample and analyse it in a laboratory, which produces accurate results but does not scale. Laboratory testing is expensive per sample, slow to return results, and impractical to repeat often enough to catch transient pollution events as they occur. Environmental agencies, municipal water authorities, and downstream communities are all affected when monitoring cannot keep pace with the rate at which pollution events appear and disappear. Delayed or missed detection can allow contamination to spread

15 before any intervention is possible, raising both the ecological cost and the
16 eventual clean-up cost. Computer vision offers a complementary angle, be-
17 cause a model that recognises the visual signature of polluted water could
18 provide a fast, low-cost first-pass screen that flags suspect locations for labo-
19 ratory follow-up. Such a screen would not replace laboratory analysis, but it
20 could help prioritise where limited testing resources are deployed. Recent re-
21 views of artificial intelligence in water monitoring confirm that image-based
22 and learning-based methods are an active and growing research direction
23 [1, 2].

24 *1.2. Machine Learning and Computer Vision Context*

25 Image-based machine learning has shifted from hand-engineered feature
26 pipelines toward end-to-end learned representations over the past decade.
27 Convolutional neural networks are well suited to visual tasks because they
28 learn relevant features directly from raw pixels rather than requiring a human
29 to design them [3]. A CNN builds a hierarchy of features, detecting simple
30 edges and colours in early layers and progressively more complex textures and
31 shapes in deeper layers. For a water-pollution task, the visually discrimina-
32 tive cues include floating debris, surface foam, discoloration, and turbidity,
33 all of which are local patterns that convolutional filters can in principle cap-
34 ture. The advantage of learned features over manual feature engineering is
35 that the network can discover combinations of cues that a human designer
36 might overlook. This flexibility comes at a cost, because CNNs are data-
37 hungry and tend to overfit when the training set is small [4]. The central
38 tension in this project is exactly that trade-off, since the available dataset
39 is far smaller than the datasets on which CNNs usually excel. Understand-
40 ing how a CNN behaves under this constraint is itself a useful outcome for
41 anyone attempting similar small-scale environmental projects.

42 *1.3. Transfer Learning for the Selected Problem*

43 Training a deep CNN from random initialisation requires a large labelled
44 dataset, which is rarely available for niche environmental tasks. Transfer
45 learning addresses this by reusing a network that has already been trained on
46 a large general dataset such as ImageNet [5]. Surveys of transfer learning for
47 image classification confirm that reusing pretrained features is an effective
48 remedy for small datasets and limited compute [6, 7, 8]. The pretrained
49 network has already learned general-purpose visual features, including edges,
50 textures, and colour gradients, that transfer across many image domains [9].

51 In a feature-extraction setup, the pretrained backbone is frozen and only
 52 a small new classification head is trained on the target data. In a fine-
 53 tuning setup, some of the later backbone layers are additionally unfrozen and
 54 adapted to the target domain. MobileNetV2 was selected here as the transfer-
 55 learning model because it is parameter-efficient and designed for resource-
 56 constrained settings [10]. Its inverted-residual structure keeps the model
 57 small while preserving representational capacity, which suits the lightweight
 58 focus of this project. A detailed architectural description of each model is
 59 deferred to the Methodology, in line with the template’s guidance.

60 *1.4. Explainable AI and Trustworthiness*

61 Relying on a single accuracy figure can be misleading, because a model
 62 may reach a given accuracy for the wrong reasons. A classifier can latch
 63 onto background cues or dataset artefacts rather than the features a human
 64 would consider meaningful, a problem often called shortcut learning. Ex-
 65 plainability methods help reveal whether a model attends to relevant image
 66 regions or to irrelevant background. Grad-CAM produces class-specific acti-
 67 vation maps that highlight which spatial regions most influenced a prediction
 68 [11]. SHAP attributes a contribution value to each feature or image region,
 69 indicating positive and negative support for a prediction [12]. For environ-
 70 mental monitoring, an unexplained or mis-focused prediction could lead to a
 71 missed pollution event or a false alarm, both of which carry real costs. This
 72 study describes how Grad-CAM and SHAP would be applied to the trained
 73 models, and it reports transparently that these explainability experiments
 74 were planned as the next phase rather than executed in the present results.
 75 This honest scoping is itself consistent with the trustworthiness argument,
 76 because overstating explainability evidence would undermine the very trust
 77 the methods are meant to build.

78 *1.5. Research Gap*

79 Although deep learning is widely applied across computer vision, com-
 80 paratively little published work targets binary clean-versus-polluted water
 81 classification using lightweight CNNs on small datasets. Most environmental-
 82 monitoring studies that use images focus on water turbidity under controlled
 83 laboratory conditions rather than on in-the-wild pollution status [13, 14].
 84 Many of these studies also rely on large, carefully curated datasets and sub-
 85 stantial computational resources, which are not available to every environ-
 86 mental agency. There is limited honest reporting of what happens when

a small, weakly labelled dataset is used, even though this is the common situation for practitioners. Reports of near-chance performance, and the reasons behind them, are under-represented because positive results are easier to publish. A further gap is the scarcity of careful data-quality auditing, including checks for label validity and data leakage, in small environmental image projects [15]. This study addresses these gaps by reporting a small-data comparison transparently, including a data-pipeline audit that uncovers a label-construction flaw. The goal is not to claim strong performance but to document the failure mode precisely enough that others can avoid or correct it.

1.6. Problem Statement

This study addresses the problem of automatically classifying water images into clean and polluted categories using a comparative and explainability-aware CNN-based framework. The images are ordinary RGB photographs of water bodies, and the prediction target is a binary label indicating pollution status. There are two nominal classes, clean water and polluted water, although the validity of these labels is itself examined in this work. The main modelling challenge is the very small dataset size, which makes overfitting likely and generalisation difficult. A secondary challenge is that some clean and polluted images are visually similar, because reflections, lighting, and surface ripples can resemble debris. The practical purpose of the system is to provide a low-cost preliminary screen that could flag suspect water bodies for laboratory follow-up rather than to replace laboratory testing. The study investigates whether transfer learning improves predictive performance over a custom CNN while keeping computation modest. It also investigates whether the reported labels actually encode the intended distinction, which turns out to be central to interpreting the results.

1.7. Aim and Objectives

The aim of this study is to develop and evaluate a lightweight, comparative, and explainability-aware CNN-based framework for classifying water images as clean or polluted, and to report its behaviour honestly on a small dataset. The specific objectives are listed below.

- analyse and preprocess the selected public water-image dataset and audit its labelling;
- develop a custom CNN as a from-scratch baseline;

- 122 • implement a MobileNetV2 transfer-learning model under identical con-
123 ditions;
- 124 • compare predictive performance using accuracy, precision, recall, F1-
125 score, and a confusion matrix;
- 126 • evaluate training behaviour, overfitting, and computational feasibility;
- 127 • describe Grad-CAM and SHAP explainability procedures and report
128 their execution status transparently;
- 129 • analyse errors, limitations, and the practical deployment suitability of
130 the approach.

131 *1.8. Research Questions*

132 The following research questions guided the experimentation and evalua-
133 tion.

- 134 • RQ1: Can a CNN trained on a small water-image dataset distinguish
135 clean water from polluted water at a useful level of accuracy?
- 136 • RQ2: Does a pretrained transfer-learning model (MobileNetV2) out-
137 perform a custom CNN trained from scratch on the same limited data?
- 138 • RQ3: How well does each model generalise to validation images it
139 did not see during training, as revealed by class-level metrics and the
140 confusion matrix?
- 141 • RQ4: What do explainability methods such as Grad-CAM and SHAP
142 contribute to interpreting the models, and what is their execution sta-
143 tus in this study?
- 144 • RQ5: What are the practical limits of this kind of image-only pollution
145 screening, including the effect of dataset size and label quality?

146 *1.9. Contributions and Novelty*

147 The novelty of this work lies not in a new architecture but in transpar-
148 ent, reproducible reporting of a small-data environmental classification ex-
149 periment, including a data-quality audit that most short reports omit. The
150 custom CNN here is intentionally simple, and the value is in the honesty
151 and completeness of the analysis around it. The explicit contributions are as
152 follows.

- 153 • a reproducible image-classification pipeline implemented in TensorFlow
154 and Keras on Google Colab, with all code openly documented;
- 155 • a controlled comparison between a from-scratch custom CNN and a
156 MobileNetV2 transfer-learning baseline under identical training condi-
157 tions;
- 158 • a data-pipeline audit that identifies a label-construction flaw, in which a
159 single unlabelled image set was randomly split into two nominal classes;
- 160 • an honest account of near-chance performance and its causes, including
161 overfitting and weak labelling, rather than an inflated success claim;
- 162 • a clearly scoped explainability plan that distinguishes methods de-
163 scribed from results actually produced.

164 The use of standard CNNs alone is not claimed as novelty, consistent with
165 the template’s guidance.

166 *1.10. Report Organization*

167 The remainder of this report is organised as follows. Section 2 reviews
168 related literature on image-based machine learning for water and environ-
169 mental monitoring, transfer-learning architectures, preprocessing, evaluation,
170 and explainability, and presents a comparative literature matrix. Section 3
171 describes the methodology in detail, including the dataset, the data-pipeline
172 audit, preprocessing, both model architectures, the training procedure, the
173 evaluation framework, and the explainability plan. Section 4 reports the
174 experimental results organised by research question, including training be-
175 haviour, overall and class-level performance, and the model comparison. Sec-
176 tion 5 discusses and interprets these results, connects them to the literature,
177 and states the study’s limitations and future directions. Section 6 concludes
178 the report.

179 **2. Literature Review**

180 *2.1. Image-Based Machine Learning in the Application Domain*

181 Image analysis for water and environmental monitoring has progressed
182 from classical image-processing techniques toward deep learning over the last
183 decade. Early approaches relied on hand-crafted descriptors of colour, tex-
184 ture, and edges, which required expert tuning and generalised poorly across

185 conditions. The success of deep convolutional networks on large benchmark
 186 datasets shifted the field toward learned representations that need little man-
 187 ual feature design [16]. Within water monitoring specifically, learning-based
 188 methods have been applied to turbidity estimation, water-quality classifi-
 189 cation, and floating-waste detection [13, 17, 18]. Related freshwater image-
 190 analysis work, such as large diatom-identification datasets, further illustrates
 191 how deep learning is reshaping ecological monitoring [19]. Interpretable deep-
 192 learning frameworks for river water quality are also emerging, signalling grow-
 193 ing attention to explainability in this domain [20]. These methods offer the
 194 practical advantage of automation and speed, but they remain sensitive to
 195 dataset size, label quality, and acquisition conditions. Reviews of the area
 196 report that water-quality monitoring is one of the most common application
 197 themes for machine learning, with classification and regression both heavily
 198 represented [1, 2]. The continuing challenge is that high accuracy is typically
 199 reported on large, controlled datasets, while small in-the-wild datasets re-
 200 main difficult. This project sits firmly in the small in-the-wild regime, which
 201 is precisely where honest reporting is most needed.

202 *2.2. CNN-Based Methods for the Selected Task*

203 CNN-based studies on water imagery have mostly addressed turbidity
 204 classification rather than binary pollution status. Nie et al. trained sev-
 205 eral CNN variants on laboratory-prepared turbidity samples and reported
 206 up to 96.5% accuracy on noisy data once a dropout layer was added [13].
 207 Soto et al. took a comparable laboratory approach but used EfficientNet-B0
 208 transfer learning on more than eleven thousand images, reaching close to 99%
 209 accuracy across five turbidity classes [14]. Feizi et al. likewise built an image-
 210 based deep learning model for laboratory turbidity estimation, again under
 211 controlled acquisition [21]. These studies share two features that the present
 212 project lacks: large datasets and tightly controlled lighting and camera con-
 213 ditions. Closer to the present setup, small-scale binary clean-versus-dirty
 214 comparisons report much more modest accuracy and show the characteristic
 215 pattern of a custom model overfitting while a pretrained model generalises
 216 slightly better. Across these works, performance is overwhelmingly dataset-
 217 limited rather than architecture-limited, which is the same conclusion the
 218 present experiments support. The comparison therefore situates this project
 219 as a small-data point that confirms the broader trend while documenting an
 220 additional labelling pitfall.

221 2.3. *Transfer Learning Architectures*

222 Several pretrained CNN architectures are commonly reused for small im-
223 age tasks. MobileNetV2 uses inverted residual blocks and linear bottlenecks
224 to remain lightweight and efficient, which suits mobile and edge deployment
225 [10]. ResNet50 introduced residual connections that allow very deep net-
226 works to train stably, and it remains a strong general-purpose backbone [22].
227 EfficientNet uses compound scaling to balance depth, width, and resolu-
228 tion, and it achieves high accuracy at modest parameter counts [23]. Recent
229 remote-sensing studies also combine such backbones in ensemble frameworks
230 for land-use and environmental image classification [24]. VGG-style networks
231 are deeper and heavier but conceptually simple, and they are often used as
232 didactic baselines [25]. For this project, MobileNetV2 was chosen because its
233 small parameter count matches the lightweight, low-resource framing of the
234 work. Its ImageNet pretraining provides general visual features that can be
235 reused without further adaptation when the backbone is frozen [5, 9]. Only
236 MobileNetV2 is treated in depth here, since it is the architecture actually
237 evaluated, while the others are noted as alternatives for future work.

238 2.4. *Data Preprocessing and Augmentation*

239 Preprocessing choices strongly affect how well a CNN learns from limited
240 data. Resizing all images to a fixed resolution and normalising pixel values
241 to a common range are standard steps that stabilise training [26]. Data aug-
242 mentation expands the effective dataset by applying label-preserving trans-
243 formations such as rotation, zoom, and horizontal flipping. Augmentation is
244 particularly valuable for small datasets because it reduces overfitting by ex-
245 posing the model to more variation [26]. However, not every transformation
246 is valid for every domain, since some operations can destroy class-relevant
247 information. For water imagery, mild rotation, zoom, and horizontal flipping
248 are reasonable, because pollution cues do not depend on a canonical orienta-
249 tion. Aggressive colour distortion would be risky, since discoloration is itself
250 a pollution cue and altering colours could remove the signal. This project ap-
251 plies modest augmentation in one configuration and none in another, which
252 allows the effect of augmentation on this small dataset to be observed.

253 2.5. *Model Evaluation and Reliability*

254 Robust evaluation requires more than a single accuracy figure, especially
255 on small or imbalanced datasets. Standard classification metrics include
256 accuracy, precision, recall, specificity, F1-score, and the area under the ROC

curve. The confusion matrix is essential because it reveals the per-class pattern of errors that a single accuracy figure hides. Several water-monitoring studies are criticised for relying on accuracy alone, which can mask poor minority-class performance. Class-level precision and recall expose whether a model is biased toward one class, which is common when data are limited. A particularly important reliability concern is data leakage, where related or duplicated images appear in both training and validation, inflating apparent accuracy [15]. This study reports accuracy, per-class precision, recall, F1-score, and the confusion matrix, and it explicitly examines whether the labels and splits are trustworthy. That examination is what separates an honest small-data report from an over-optimistic one.

2.6. Explainable AI in Image Analysis

2.6.1. Grad-CAM

Grad-CAM produces class-specific activation maps by using the gradients of a target class flowing into the final convolutional layer [11]. The resulting heatmap highlights the image regions that most increased the score for the predicted class. In related image studies it is used to check whether a model attends to the object of interest rather than to background. Its main value is diagnostic, because it can reveal shortcut learning that accuracy alone would hide. Its main limitation is coarse spatial resolution, since the final convolutional layer has a small spatial footprint. For this project, Grad-CAM would be applied to the trained custom CNN to inspect whether attention falls on water-surface debris and discoloration.

2.6.2. SHAP

SHAP assigns each feature or image region a contribution value based on cooperative game theory, indicating positive and negative support for a prediction [12]. For images it typically operates over superpixels or masked regions, estimating how each region changes the model output. This produces a fine-grained map of which regions push the prediction toward clean or toward polluted. SHAP is more computationally expensive than Grad-CAM because it requires many masked evaluations per image. Its interpretation also requires care, since attribution values depend on the chosen background or reference images. For this project, SHAP would complement Grad-CAM by providing region-level positive and negative evidence on the same example images.

292 2.6.3. *Explainability Limitations*

293 An attractive heatmap is not proof of correct reasoning, and explain-
294 ability outputs must be interpreted critically. Explanations can be sensitive
295 to preprocessing choices and can shift noticeably with small input changes.
296 They are also model-dependent, so two models with similar accuracy may
297 attend to quite different regions. A heatmap may highlight a region that cor-
298 relates with the label without being causally meaningful. Domain knowledge
299 is therefore needed to judge whether the highlighted regions are genuinely
300 pollution-relevant. Because of these caveats, this report treats explainabil-
301 ity as a diagnostic aid rather than as definitive evidence, and it does not
302 overstate explainability results that were not produced.

303 2.7. *Robustness, Generalization, and Deployment*

304 Real-world deployment exposes a model to image noise, blur, varying illu-
305 mination, occlusion, and domain shift. Studies that target deployment often
306 evaluate performance under such degradations to estimate field reliability.
307 Lightweight architectures such as MobileNetV2 are attractive for deployment
308 because of their small size and fast inference [10]. Robustness testing was
309 outside the scope of the present results, because the more fundamental data
310 and labelling problems had to be addressed first. Computational feasibility,
311 however, is reviewed, since both models train quickly on a single Colab GPU.
312 The honest position is that deployment is premature for these models, and
313 robustness evaluation belongs to a later phase once a valid dataset is in place.

314 2.8. *Comparative Literature Review Matrix*

315 Table 1 compares closely related image-based water and environmental
316 classification studies across datasets, models, evaluation practices, explain-
317 ability, robustness, validation, and reported limitations. The matrix is in-
318 tended to synthesise the field rather than to restate each paper, and the
319 final row positions the present study against the reviewed work. The com-
320 parison shows several recurring patterns that motivate this project. Most
321 reviewed studies use controlled or large datasets and report high accuracy,
322 while explainability, robustness testing, and external validation are incon-
323 sistently applied. Class-imbalance treatment and statistical testing are fre-
324 quently absent, and very few studies audit label validity or data leakage.
325 The present study deliberately occupies the under-reported corner of small
326 datasets with weak labels, and its contribution is the transparent documen-
327 tation of that regime. The synthesis that follows the table discusses these

328 common strengths, recurring weaknesses, and missing evaluations in more
329 depth.

Table 1: Comparative literature review matrix summarising datasets, models, evaluation practices, explainability methods, robustness analysis, and validation strategies across closely related image-based water and environmental classification studies, with the proposed study shown in the final row. Symbols: ✓ included, × not included, ∼ partial or unclear. Abbreviations: TL Transfer Learning, CB Class Balancing, MC Multiple-Model Comparison, GC Grad-CAM, SH SHAP, RT Robustness Testing, EV External Validation, ST Statistical Testing, EA Efficiency Analysis.

Study, Year	Problem	Dataset	Cls	Size	Models	TL	CB	MC	GC	SH
Nie et al., 2025 [13]	Turbidity classification	Lab turbidity	5	Lab set	CNN variants	×	∼	✓	×	×
Soto et al., 2025 [14]	Turbidity classification	Lab RGB	5	11k+	EfficientNet-B0	✓	×	×	×	×
Feizi et al., 2023 [21]	Turbidity estimation	Lab images	—	Lab set	Deep CNN	∼	×	×	×	×
Nunkhaw & Miyamoto, 2024 [18]	Floating waste	River images	Multi	Field set	Deep learning	✓	×	∼	×	×
Chatterjee et al., 2024 [17]	WQ classification	Tabular WQ	Multi	523	Classic ML	×	∼	✓	×	×
Boucetta et al., 2024 [27]	WQ assessment	Sensor data	Multi	523	DNN	×	×	✓	×	×
Parra et al., 2024 [28]	Turbidity quantify	RGB sensor	Multi	Lab set	ML + RGB	×	×	✓	×	×
Proposed, 2025	Clean vs polluted	Kaggle water	2	293	Custom CNN, MobileNetV2	✓	×	✓	∼	∼

330 2.9. Synthesis of Literature and Identified Gap

331 Synthesising the reviewed studies, the field does several things consis-
332 tently well and a few things poorly. It reliably demonstrates that CNNs and
333 transfer learning can classify water imagery accurately when datasets are
334 large and acquisition is controlled [13, 14, 21]. It consistently favours trans-
335 fer learning for small or moderate datasets, which matches the rationale for
336 including MobileNetV2 here [10, 9]. Where studies disagree is on whether
337 lightweight or heavy backbones are preferable, since this depends on the
338 dataset and the deployment constraints. What is methodologically missing
339 across much of the small-scale literature is rigorous data-quality auditing,
340 explainability validation, and honest reporting of near-chance outcomes [15].
341 External validation and statistical testing are also rare, which limits how
342 far reported accuracies can be trusted. The present study responds to these

gaps by auditing the data pipeline, reporting near-chance results transparently, and clearly scoping which explainability experiments were executed. This sets up the Methodology, which describes exactly how the data were processed, how the models were built and trained, and how they were evaluated.

3. Methodology

The methodology describes the full experimental design in enough detail for another researcher to reproduce the study. It is experimental and quantitative, the task is binary image classification, and two models are compared under identical conditions.

3.1. Research Design

This work is an experimental, quantitative, comparative image-classification study. Two models are evaluated: a custom CNN trained from scratch and a MobileNetV2 transfer-learning model with a frozen backbone. Both models are trained and validated on the same data split so that the comparison is fair. Explainability via Grad-CAM and SHAP is included in the design as a planned diagnostic stage, with execution status reported honestly. The evaluation strategy combines training-behaviour analysis, overall accuracy, class-level metrics, and confusion-matrix inspection. The overall workflow proceeds from dataset selection and inspection, through cleaning and preprocessing, data splitting, optional augmentation, model development, training and validation, evaluation, and practical interpretation. Figure 1 summarises this end-to-end pipeline. The design deliberately includes a data-pipeline audit step, which is what surfaces the labelling problem central to the results.

3.2. Dataset Description

3.2.1. Dataset Source and Accessibility

The dataset is a publicly available water-image set obtained from a Kaggle repository. It was accessed and downloaded for this project and used under its public-dataset terms. The dataset was selected because it is small, openly available, and representative of the low-resource setting this project targets. The source archive contained a single folder of water images without a pre-existing clean-versus-polluted directory structure. This single-folder organisation is important, because the class split was created during preprocessing rather than supplied by the dataset. The dataset link is stored in a text file in the Datasets folder of the repository, as required by the template.

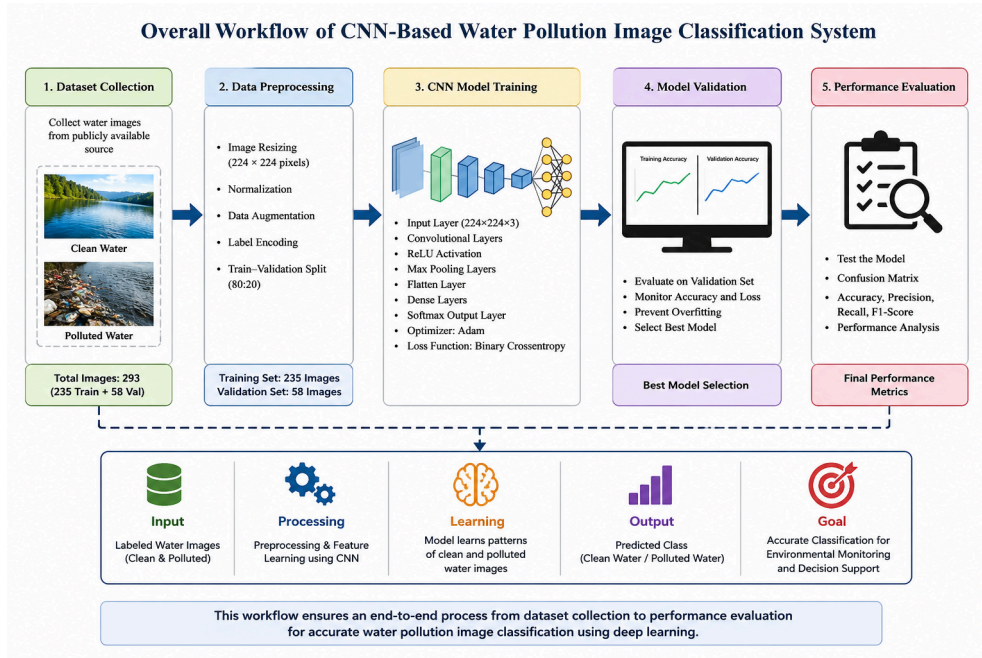


Figure 1: Overall workflow of the CNN-based water pollution image classification system, showing dataset collection, preprocessing and augmentation, custom CNN and MobileNetV2 model construction, training and validation, performance evaluation, and the planned explainability stage. The pipeline is the same for both models so that the comparison is conducted under identical conditions.

3.2.2. Dataset Composition

The source archive contained 293 water images in mixed formats, predominantly JPG with some PNG, JPEG, and WEBP files. After organisation and cleaning, the working dataset comprised 234 training images and 58 validation images. All images were standardised to a resolution of 224×224 pixels with three RGB channels before training. The images vary widely in size, lighting, viewpoint, and content, which is typical of in-the-wild photographs. One corrupted image was detected and removed during cleaning, as described below. The images carry no expert annotation; the only labels are those constructed during preprocessing. Table 2 summarises the dataset configuration.

3.2.3. Class Definitions

The two nominal classes are clean water and polluted water. Conceptually, polluted-water images should show visible contamination such as floating plastic, industrial debris, foam, or discoloured surfaces. Clean-water images should show relatively clear rivers, lakes, or coastal water with minimal visible pollution. In practice, however, the labels in this dataset were not assigned according to these definitions, as the audit in the next subsection explains. This gap between the intended class definitions and the actual label assignment is the most important methodological finding of the study. It is stated here explicitly so that the class definitions are not mistaken for validated ground truth.

3.2.4. Dataset Quality Assessment and Pipeline Audit

A careful audit of the data pipeline revealed a critical label-construction flaw. The preprocessing code took the single folder of 293 unlabelled images, shuffled it randomly, and split it in half, copying one half into a folder named “clean” and the other half into a folder named “polluted”. This means the class labels were assigned at random and do not encode any real clean-versus-polluted distinction. As a direct consequence, no model can learn a meaningful decision boundary, because the labels carry essentially no signal about pollution status. This single fact explains why both models perform near the 50% chance level on validation data. Beyond the labelling flaw, the dataset is small, visually heterogeneous, and unverified against expert annotation, all of which further limit reliability. There is also a risk of subtle data leakage and class imbalance that cannot be ruled out given the construction method [15]. Reporting this audit transparently is the central contribution

414 of the study, because it converts an apparent modelling failure into a clear,
415 actionable data-quality lesson.

416 3.2.5. Dataset Visualization

417 Representative clean-water and polluted-water example images are shown
418 in Figures 2 and 3. These examples illustrate the visual variety in the dataset,
419 including different water bodies, lighting, and viewpoints. It should be read
420 with the audit in mind, since the folder names reflect the random split rather
421 than verified content. The visual heterogeneity also helps explain why a small
422 model struggles, independent of the labelling issue.

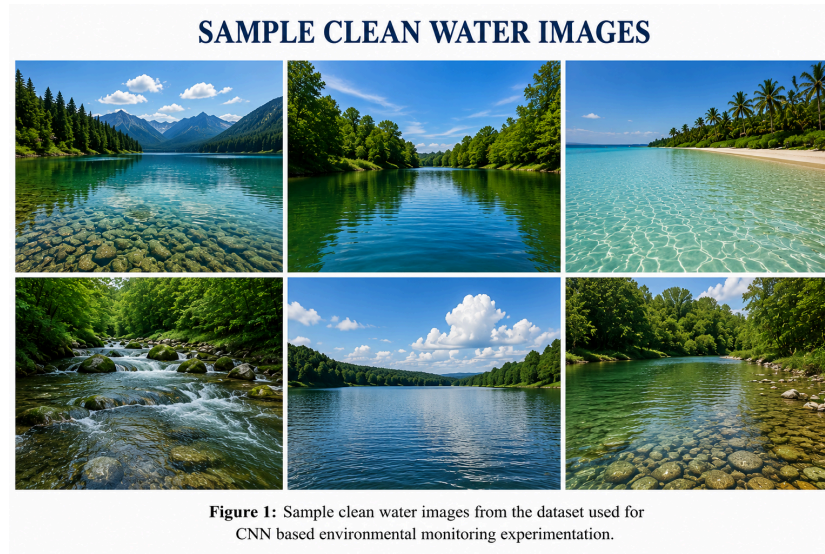


Figure 1: Sample clean water images from the dataset used for CNN based environmental monitoring experimentation.

Figure 2: Representative sample images from the folder labelled clean water, illustrating the range of water bodies, lighting conditions, and viewpoints present in the dataset. The folder label reflects the random preprocessing split rather than verified clean-water content, as explained in the dataset quality assessment.

423 3.3. Data Preparation

424 3.3.1. Data Cleaning

425 Data cleaning consisted of verifying that each image file could be opened
426 and decoded. Each file was opened with the Pillow library and verified,
427 and any file that failed verification was removed. This process detected and
428 removed one corrupted image from the polluted folder, leaving 292 usable

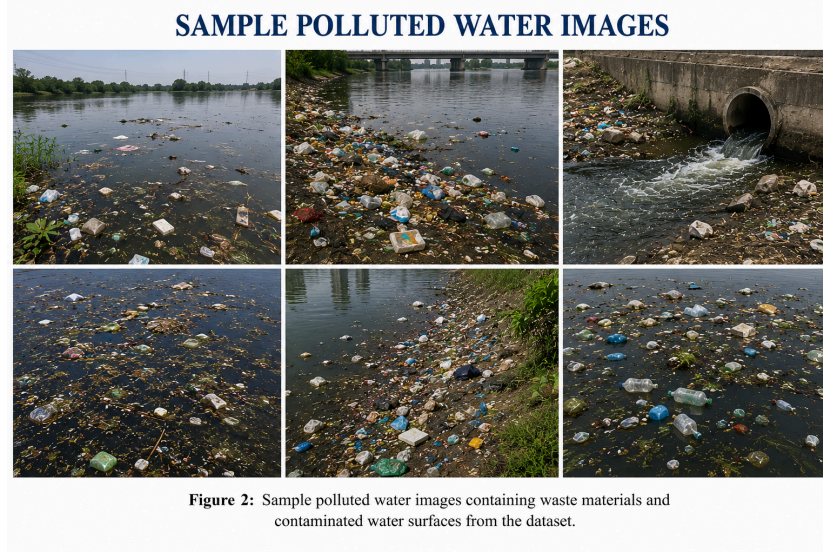


Figure 3: Representative sample images from the folder labelled polluted water, showing contaminated surfaces, floating waste, and discoloration alongside visually ambiguous cases. As with the clean-water folder, the label reflects the random split rather than verified pollution status.

Table 2: Dataset description summary used for CNN-based water pollution image classification, listing the dataset type, classes, image counts, resolution, preprocessing operations, split ratio, and implementation environment used during model training and validation.

Dataset Property	Description
Dataset Type	Water images (public Kaggle set)
Classification Categories	Clean Water, Polluted Water (randomly assigned)
Source Images	293
Training Images	234
Validation Images	58
Image Resolution	224×224 pixels, RGB
Data Source	Public online dataset
Preprocessing	Resizing, normalisation, optional augmentation
Data Split Ratio	80:20 (training:validation)
Implementation Platform	Google Colab (GPU)
Deep Learning Framework	TensorFlow / Keras

429 images. No duplicate-detection or label-verification step was applied beyond
430 this, which is itself a limitation given the labelling flaw. Unreadable files
431 were excluded rather than repaired, since repair was not feasible for corrupted
432 content. The cleaning step is simple but necessary, because a single corrupted
433 file can interrupt training.

434 3.3.2. *Data Splitting*

435 The data were split into training and validation subsets using an 80:20
436 ratio. This produced 234 training images and 58 validation images. The split
437 was performed by the Keras ImageDataGenerator validation-split mechanism
438 applied to the organised class folders. Because the class folders were them-
439 selves created by a random half-split of the source images, the training and
440 validation sets inherit randomly assigned labels. No separate independent
441 test set was held out, which is a limitation for estimating true generalisation.
442 Stratification was implicit in the generator, but it cannot compensate for
443 labels that carry no real signal.

444 3.3.3. *Image Preprocessing*

445 All images were resized to 224×224 pixels to provide a consistent input
446 size for both models. Pixel values were rescaled from the 0–255 range to the
447 0–1 range by dividing by 255, which stabilises optimisation. Images were
448 kept in three-channel RGB form, since colour is a potentially relevant cue
449 for pollution. The 224×224 size was chosen because it matches the input ex-
450 pected by MobileNetV2’s ImageNet pretraining, allowing a fair comparison.
451 No cropping or contrast enhancement was applied, in order to keep prepro-
452 cessing simple and reproducible. Each operation is justified by the need for
453 consistent, normalised inputs that suit both the custom CNN and the pre-
454 trained backbone. Figure 4 illustrates the preprocessing pipeline from the
455 raw image through resizing and normalisation to the final CNN input tensor.

456 3.3.4. *Data Augmentation*

457 Augmentation was applied in one training configuration to expand the
458 effective dataset and reduce overfitting [26]. The augmentation comprised
459 random rotation up to twenty degrees, zoom up to twenty percent, and hor-
460 izontal flipping. These transformations are label-preserving for water im-
461 agery, because pollution cues do not depend on a fixed orientation. A second
462 configuration used only rescaling without augmentation, which allows the
463 effect of augmentation to be observed. Colour-altering augmentations were

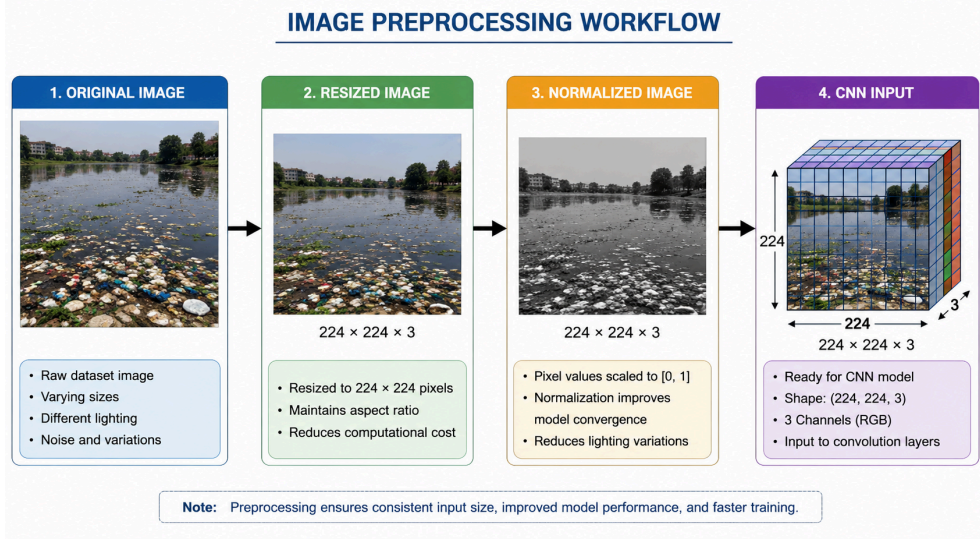


Figure 4: Image preprocessing workflow showing the transformation of a raw water image into a model-ready input, comprising resizing to 224×224 pixels, pixel-value normalisation to the 0–1 range, and assembly into a $224 \times 224 \times 3$ RGB tensor. Consistent preprocessing stabilises optimisation and ensures both models receive identical input formats.

deliberately avoided, since discoloration is a pollution cue that such transformations could corrupt. The modest augmentation parameters reflect the small dataset and the desire not to introduce unrealistic distortions.

3.3.5. Class-Imbalance Handling

No explicit class-balancing method such as class weights, oversampling, or focal loss was applied. Because the two folders were created by an even random half-split, the nominal class sizes are approximately balanced. However, nominal balance is meaningless here, since the labels do not correspond to real content. The decision not to apply balancing is reasonable given the approximate count balance, but it does not address the deeper labelling problem. This is noted as a limitation rather than presented as a solved issue.

3.4. Model Development

3.4.1. Custom CNN Baseline

The custom CNN takes a $224 \times 224 \times 3$ input and applies three convolutional blocks for hierarchical feature extraction. The first block uses 32 filters,

the second 64 filters, and the third 128 filters, each with a 3×3 kernel and ReLU activation. Each convolutional layer is followed by a 2×2 max-pooling layer that reduces spatial dimensions and computation. The feature maps are then flattened into a one-dimensional vector and passed to a dense layer of 128 units with ReLU activation. A dropout layer with a rate of 0.5 is applied to reduce overfitting [29]. The output layer is a single unit with a sigmoid activation that produces a probability for the binary classification. The model is compiled with the Adam optimiser and the binary cross-entropy loss [30]. Figure 5 shows the full architecture, and Table 3 lists the hyperparameters.

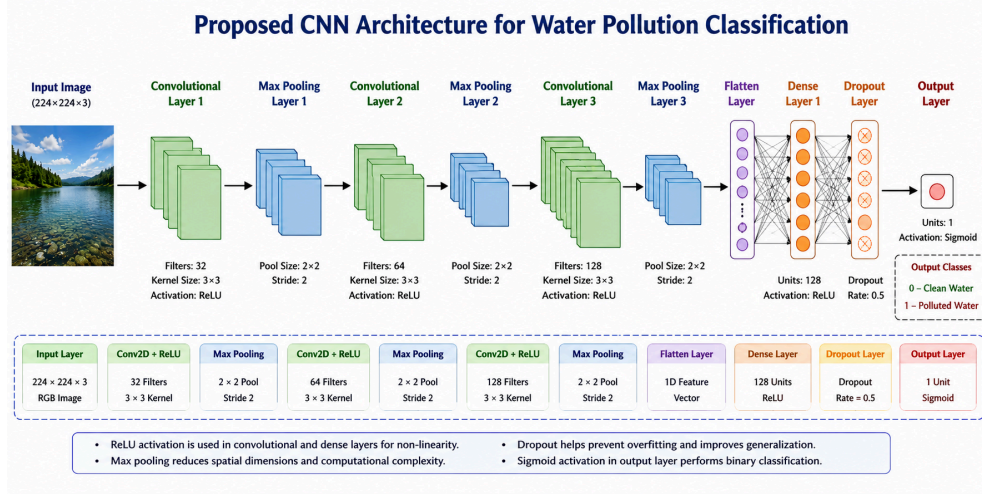


Figure 5: Architecture of the proposed custom convolutional neural network, showing three convolutional blocks with 32, 64, and 128 filters, each followed by max pooling, a flatten layer, a 128-unit dense layer, a dropout layer at rate 0.5, and a single sigmoid output unit for binary clean-versus-polluted classification.

3.4.2. Transfer-Learning Model One: MobileNetV2

The transfer-learning model uses MobileNetV2 pretrained on ImageNet as a frozen feature extractor [10, 5]. The original ImageNet classification head is removed, leaving the convolutional backbone. The backbone weights are frozen so that ImageNet-derived features are reused without modification. A global average pooling layer follows the backbone to summarise each feature map into a single value. A new dense output layer with a single sigmoid unit is added for binary classification. The model is compiled with the Adam optimiser and binary cross-entropy loss, matching the custom CNN. Because

the backbone is frozen, only the small new head is trained, which suits the very small dataset.

3.4.3. Feature Extraction and Fine-Tuning Strategy

This study uses the feature-extraction stage of transfer learning, with the MobileNetV2 backbone fully frozen. No fine-tuning of backbone layers was performed, so the reported MobileNetV2 result reflects frozen-feature transfer only. This choice was deliberate, because fine-tuning a large backbone on 234 images would risk severe overfitting. It also means the reported MobileNetV2 performance likely understates what a carefully fine-tuned model could achieve on a valid dataset. Which layers to unfreeze, and in what order, is left to future work once the labelling problem is corrected.

3.5. Training Procedure

3.5.1. Loss Function

Both models use binary cross-entropy loss, which is the standard choice for two-class problems with a sigmoid output. Binary cross-entropy measures the divergence between the predicted probability and the true binary label. It is appropriate here because each image is assigned to one of exactly two nominal categories. The same loss is used for both models so that the comparison is not confounded by differing objectives.

3.5.2. Optimizer and Learning Rate

The Adam optimiser was used for both models because of its adaptive per-parameter learning rates and robust default behaviour [30]. The default Keras Adam learning rate of 0.001 was used, and no explicit learning-rate schedule was applied. No additional weight decay or L2 regularisation was added beyond dropout in the custom CNN. These settings were chosen for simplicity and reproducibility on a small dataset.

3.5.3. Training Controls

Both models were trained for five epochs with a batch size of 32. No early stopping or model checkpointing was used, given the short five-epoch schedule. Training ran on the GPU provided by the Google Colab environment. The short schedule was a reasonable choice for the dataset size, although it limits the analysis of long-run convergence. Random shuffling was applied by the data generator, but a fixed random seed was not set, so exact runs may vary slightly.

532 3.5.4. Hyperparameter Selection

533 Hyperparameters were selected from common literature defaults and con-
534 strained by the small dataset and modest compute. The filter counts, dropout
535 rate, optimiser, and batch size follow widely used conventions for small CNNs.
536 No systematic hyperparameter search was performed, which is acknowledged
537 as a limitation. Table 3 lists the complete set of hyperparameters used for
538 both models.

Table 3: Hyperparameter configuration used for both the custom CNN and the MobileNetV2 transfer-learning model, listing input size, optimiser, learning rate, loss function, batch size, epochs, dropout, and the training environment. Shared settings ensure a controlled comparison between the two models.

Hyperparameter	Value
Input image size	$224 \times 224 \times 3$
Optimiser	Adam
Learning rate	0.001 (Keras default)
Loss function	Binary cross-entropy
Batch size	32
Epochs	5
Custom CNN dropout rate	0.5
MobileNetV2 backbone	Frozen (ImageNet weights)
Activation (hidden)	ReLU
Activation (output)	Sigmoid
Framework	TensorFlow / Keras
Environment	Google Colab (GPU)

539 3.6. Evaluation Framework

540 3.6.1. Primary Predictive Metrics

541 Model performance is evaluated using accuracy, precision, recall, and F1-
542 score, defined specifically for this clean-versus-polluted task. Accuracy is the
543 fraction of validation water images whose predicted pollution label matches
544 the assigned label, as in Equation 1. Precision for the polluted class is the
545 fraction of images predicted polluted that are actually labelled polluted, as
546 in Equation 2. Recall for the polluted class is the fraction of truly polluted-
547 labelled images that the model correctly identifies, as in Equation 3. The
548 F1-score is the harmonic mean of precision and recall, balancing the two, as

549 in Equation 4. These metrics are reported per class and as averages, because
 550 a single accuracy figure would hide the per-class behaviour.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (1)$$

$$\text{Precision} = \frac{TP}{TP + FP} \quad (2)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (3)$$

$$\text{F1} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \quad (4)$$

551 Here TP , TN , FP , and FN denote true positives, true negatives, false
 552 positives, and false negatives for the polluted class on the validation set.

553 3.6.2. Class-Level Evaluation

554 Class-level evaluation reports per-class precision, recall, and F1-score for
 555 both the clean and polluted classes. The confusion matrix is used to expose
 556 the exact pattern of true and false predictions per class. Particular attention
 557 is paid to whether the model systematically favours one class, which is com-
 558 mon when labels carry little signal. This level of detail is what reveals that
 559 the models are effectively guessing rather than discriminating.

560 3.6.3. Training Behaviour

561 Training behaviour is evaluated by tracking training and validation accu-
 562 racy and loss across the five epochs. The gap between training and validation
 563 curves is used to diagnose overfitting. Convergence and stability are assessed
 564 qualitatively from the shape of the curves. This analysis distinguishes the
 565 custom CNN’s overfitting pattern from MobileNetV2’s flatter, near-chance
 566 behaviour.

567 3.6.4. Computational Efficiency

568 Computational feasibility is assessed qualitatively, since both models train
 569 within minutes on a single Colab GPU. The custom CNN is small, with three
 570 convolutional blocks and one dense layer, so it trains quickly. MobileNetV2
 571 is larger but still lightweight, and with a frozen backbone only the small
 572 head is trained. Detailed parameter counts and inference timings were not
 573 the focus of this phase and are noted for future measurement.

574 3.6.5. *Statistical Reliability*

575 No repeated runs, confidence intervals, or significance tests were com-
576 puted in this phase. Given that both models perform near chance for an
577 identified labelling reason, formal statistical comparison would be prema-
578 ture. Differences of a few percentage points between the two models are
579 therefore not claimed to be statistically meaningful. Repeated runs and con-
580 fidence intervals are recommended for future work once a valid dataset is
581 available.

582 3.7. *Explainable AI Methodology*

583 3.7.1. *Grad-CAM Procedure*

584 The planned Grad-CAM procedure would target the final convolutional
585 layer of the custom CNN. For a given image, the gradient of the predicted-
586 class score with respect to that layer’s feature maps would be used to weight
587 the maps. The weighted maps would be combined and passed through a
588 ReLU to form a class-specific heatmap. The heatmap would be normalised
589 and overlaid on the original image to show which regions drove the prediction.
590 Example images would be selected to cover correct and incorrect predictions
591 across both classes. This procedure is described for completeness, and its
592 execution status is reported transparently in the Results.

593 3.7.2. *SHAP Procedure*

594 The planned SHAP procedure would use an image masking explainer with
595 a small set of background reference images. The image would be partitioned
596 into regions, and many masked evaluations would estimate each region’s con-
597 tribution. Positive and negative attributions would indicate which regions
598 push the prediction toward polluted or toward clean. The same example
599 images used for Grad-CAM would be explained with SHAP to allow direct
600 comparison. As with Grad-CAM, this is presented as a described procedure
601 whose execution status is reported honestly.

602 3.7.3. *Explanation Assessment*

603 Explanations would be judged by whether attention falls on genuinely
604 pollution-relevant regions such as debris, foam, or discoloration. Agreement
605 between Grad-CAM and SHAP on the same image would increase confidence
606 in the diagnosis. Differences between explanations for correct and incorrect
607 predictions would be examined for shortcut learning. Crucially, because the

608 labels in this dataset are randomly assigned, any explanation would be ex-
609 pected to look unstable or meaningless, which is itself a useful confirmation
610 of the labelling problem.

611 *3.8. Software, Hardware, and Reproducibility*

612 The experiments were implemented in Python using TensorFlow and
613 Keras within Google Colab. Training used the GPU provided by the Colab
614 runtime. The main libraries were TensorFlow/Keras for modelling, scikit-
615 learn for metrics, Pillow for image handling, and Matplotlib for plotting. All
616 code is documented in the accompanying notebook and stored in the Codes
617 folder of the repository, with the GitHub link recorded in a text file. A fixed
618 random seed was not set, so minor run-to-run variation is expected. Saved
619 outputs, including accuracy and loss curves and the confusion matrix, are
620 stored alongside the code.

621 *3.9. Ethical and Practical Considerations*

622 The most important ethical consideration here is not to overstate the
623 system’s capability. Because the models perform at chance for an identified
624 labelling reason, presenting them as a working pollution detector would be
625 misleading and potentially harmful. An over-trusted but unreliable envi-
626 ronmental screen could divert testing resources or miss real contamination.
627 The dataset contains no personal data, so privacy and consent concerns are
628 minimal. Any future deployment of such a system would require human ver-
629 ification of all flagged cases. This study summarises its methodology in one
630 workflow figure and emphasises that the honest reporting of failure is the
631 responsible course.

632 *3.10. Dataset*

633 The dataset used in this study is the publicly available Kaggle water-
634 image set described in Section 3.2, comprising 293 source images standardised
635 to 234 training and 58 validation images at 224×224 resolution. The single
636 most important property of this dataset, established by the pipeline audit,
637 is that its clean and polluted folders were produced by a random half-split
638 of unlabelled images rather than by content-based labelling. This property
639 is summarised in Table 2 and is the key to interpreting every result that
640 follows. The dataset link is stored in a text file in the Datasets folder of the
641 repository, as required.

642 3.11. Detailed Methodology

643 The complete pipeline can be summarised in three stages that map onto
 644 Figure 1. In the first stage, the 293 source images are organised into two
 645 folders, one corrupted file is removed, and all images are resized to 224×224
 646 and normalised to the 0–1 range. In the second stage, the data are split
 647 80:20 into 234 training and 58 validation images, optionally augmented with
 648 rotation, zoom, and horizontal flipping, and fed to the two models. In the
 649 third stage, the custom CNN and the frozen-backbone MobileNetV2 are each
 650 trained for five epochs with Adam and binary cross-entropy, after which
 651 accuracy, loss, per-class metrics, and the confusion matrix are computed.
 652 The workflow is identical for both models so that any performance difference
 653 reflects the model rather than the pipeline. The convolution operation that
 654 underlies feature extraction is given in Equation 5, where I is the input
 655 image, K is the kernel, and F is the resulting feature map.

$$F(i, j) = \sum_m \sum_n I(i - m, j - n) \cdot K(m, n) \quad (5)$$

656 The ReLU activation used in the hidden layers is given in Equation 6,
 657 and the sigmoid activation used in the output layer is given in Equation 7.

$$f(x) = \max(0, x) \quad (6)$$

$$\sigma(x) = \frac{1}{1 + e^{-x}} \quad (7)$$

658 3.12. Experimental settings

659 The experimental settings are identical for the two competing models
 660 except for the architecture itself. Both use $224 \times 224 \times 3$ inputs, the Adam
 661 optimiser, binary cross-entropy loss, a batch size of 32, and five training
 662 epochs in Google Colab. The custom CNN adds a dropout layer at rate 0.5,
 663 while MobileNetV2 uses a frozen ImageNet backbone with a single new sig-
 664 moid output unit. The full hyperparameter configuration is given in Table 3,
 665 and the custom CNN architecture is shown in Figure 5.

666 4. Results

667 The results are reported without extended interpretation and are organ-
 668 ised by research question. Detailed interpretation is reserved for the Discus-
 669 sion.

670 4.1. Dataset and Preprocessing Results

671 After cleaning, the dataset comprised 234 training and 58 validation im-
672 ages at 224×224 resolution, with one corrupted image removed from the orig-
673 inal 293. The data generator confirmed 234 training images and 58 validation
674 images across the two nominal classes. The class folders were approximately
675 balanced by construction, because the source images were split in half. The
676 preprocessing produced consistent normalised inputs for both models. The
677 most consequential preprocessing result, however, is the audit finding that
678 the class labels were assigned randomly rather than by content. This finding
679 directly conditions the interpretation of every subsequent result and relates
680 to RQ1 and RQ5.

681 4.2. Training and Convergence Results

682 The custom CNN’s training accuracy rose steadily from about 62% in
683 the first epoch to 79.06% in the fifth epoch. Its validation accuracy, by con-
684 trast, stayed in the 40–48% range and ended at 43.10%, showing a widening
685 train–validation gap. Its training loss fell from about 0.66 to 0.48, while its
686 validation loss rose from about 0.72 to 0.84. This diverging pattern, with
687 falling training loss and rising validation loss, is the classic signature of over-
688 fitting. MobileNetV2’s training accuracy rose more gently from about 48%
689 to 67.5%, while its validation accuracy stayed near 47–52% and ended at
690 50.00%. MobileNetV2’s curves are flatter and show less overfitting but no
691 real learning of a meaningful signal. Figures 6 and 7 present the custom CNN
692 accuracy and loss curves. These training-behaviour results relate to RQ3.

693 4.3. Overall Model Performance

694 Table 4 compares the custom CNN and MobileNetV2 on training and val-
695 idation accuracy. The custom CNN reached the higher training accuracy at
696 79.06%, but the lower validation accuracy at 43.10%. MobileNetV2 reached
697 a lower training accuracy of about 67.5% but a higher validation accuracy
698 of 50.00%. Numerically, MobileNetV2 is the stronger model on validation
699 accuracy, though both sit at or below the 50% chance level for a binary task.
700 No model achieves performance that would be useful in practice, which is
701 consistent with the labelling audit. This overall comparison relates to RQ2.

702 4.4. Class-Level Performance

703 The confusion matrix for the custom CNN on the 58 validation images
704 was $\begin{bmatrix} 15 & 14 \\ 17 & 12 \end{bmatrix}$, shown in Figure 8. For the clean class (label 0)

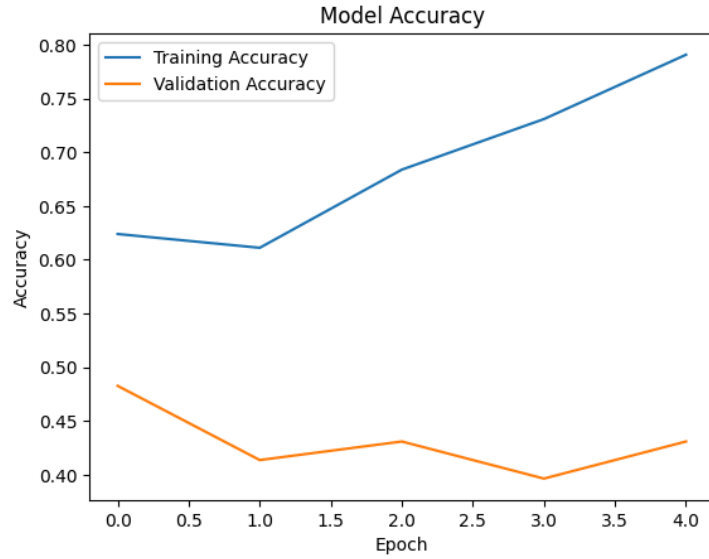


Figure 6: Training and validation accuracy curves of the custom convolutional neural network across five epochs. Training accuracy rises to 79.06% while validation accuracy remains near chance and ends at 43.10%, illustrating pronounced overfitting on the small dataset.

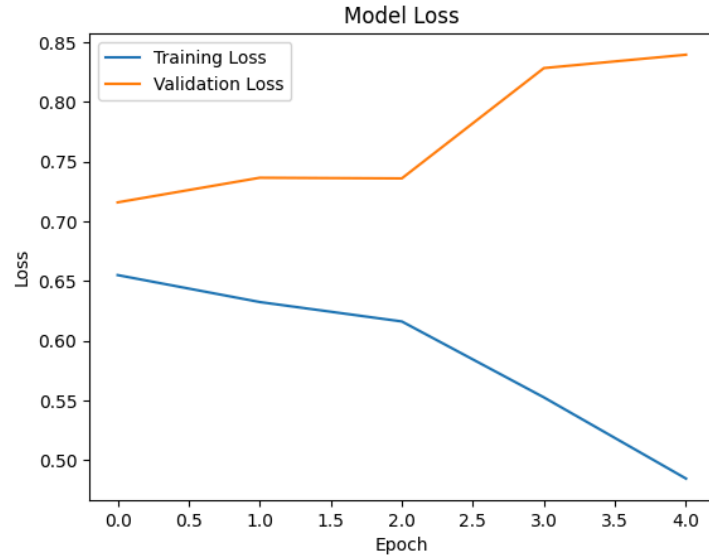


Figure 7: Training and validation loss curves of the custom convolutional neural network across five epochs. Training loss decreases steadily while validation loss increases, the characteristic divergence that confirms overfitting rather than genuine generalisation.

Table 4: Comparison of the custom CNN and the MobileNetV2 transfer-learning model on training and validation accuracy under identical training conditions. The custom CNN achieves higher training accuracy but lower validation accuracy, while MobileNetV2 generalises marginally better; both remain at or below the binary chance level.

Model	Training Accuracy (%)	Validation Accuracy (%)
Custom CNN	79.06	43.10
MobileNetV2 (frozen)	67.52	50.00

705 the model achieved precision 0.47, recall 0.52, and F1-score 0.49, and for
 706 the polluted class (label 1) it achieved precision 0.46, recall 0.41, and F1-
 707 score 0.44. The overall validation accuracy from the classification report
 708 was 0.47, with macro-averaged precision, recall, and F1 all near 0.46–0.47.
 709 These near-equal, near-chance per-class values indicate that the model is
 710 effectively guessing rather than discriminating. The errors are distributed
 711 roughly evenly across both classes, with no class reliably recognised. Table 5
 712 reports the full classification report. This class-level analysis relates to RQ3.

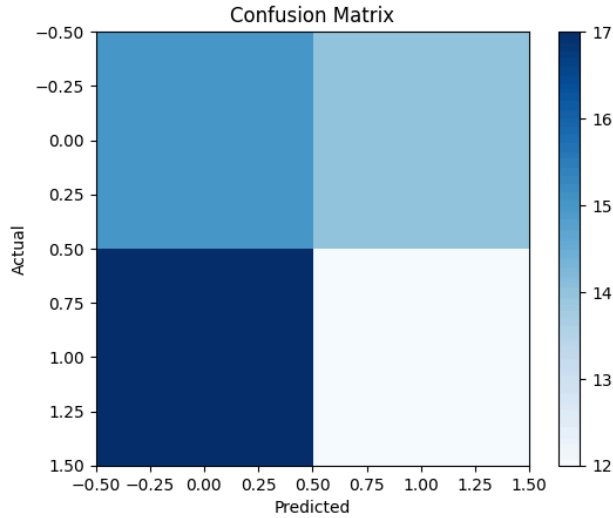


Figure 8: Confusion matrix of the custom convolutional neural network on the 58 validation images, with values $[[15, 14], [17, 12]]$ for clean and polluted classes. The near-uniform distribution of correct and incorrect predictions across both classes reflects near-chance, undiscriminating behaviour.

Table 5: Classification report for the custom CNN on the validation set, giving per-class precision, recall, and F1-score together with the macro average. The near-equal, near-0.46 values across both classes confirm that the model does not discriminate between the two nominal categories.

Class	Precision	Recall	F1-score
Clean Water (0)	0.47	0.52	0.49
Polluted Water (1)	0.46	0.41	0.44
Macro Average	0.47	0.47	0.46

713 4.5. Explainable AI Results

714 Grad-CAM and SHAP were specified in the methodology as the planned
 715 explainability stage for this project. These experiments were not executed
 716 in the present phase, because the data-pipeline audit revealed that the la-
 717 bels are randomly assigned, which makes any explanation uninterpretable
 718 on the current dataset. Running Grad-CAM or SHAP on a model trained
 719 against random labels would highlight regions with no meaningful relation-
 720 ship to pollution, so producing such figures would be misleading rather than
 721 informative. The honest result for this subsection is therefore that explain-
 722 ability is deferred until a validly labelled dataset is available, at which point
 723 the procedures in Section 3.8 will be applied. This transparent reporting is
 724 preferable to presenting fabricated or meaningless heatmaps, and it relates
 725 directly to RQ4.

726 4.6. Efficiency and Deployment Results

727 Both models trained within a few minutes on a single Colab GPU, con-
 728 firming that the approach is computationally feasible. The custom CNN
 729 is compact, and MobileNetV2 with a frozen backbone trains only a small
 730 head, so both are lightweight. However, computational feasibility does not
 731 imply deployment readiness, because neither model performs above chance.
 732 No model is suitable for deployment in its current state, and deployment
 733 is therefore not recommended. This efficiency and deployment assessment
 734 relates to RQ5.

735 For qualitative illustration, Figure 9 shows two individual high-confidence
 736 predictions produced by the custom CNN. These two hand-picked examples
 737 happen to be classified correctly with high confidence, but they are illustra-
 738 tive single cases only and must not be read as representative of overall per-
 739 formance. The aggregate validation accuracy of 43.10% and the near-chance

740 confusion matrix in Figure 8 are the authoritative measures, and they show
 741 that such confident correct predictions do not generalise across the validation
 742 set.

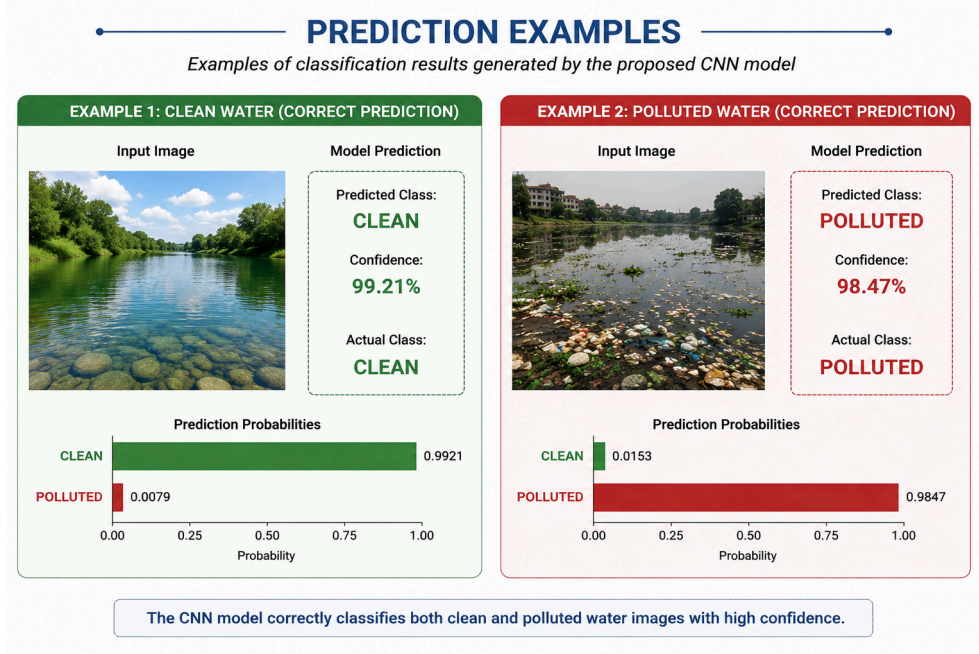


Figure 9: Two illustrative single-image predictions from the custom CNN, each classified correctly with high confidence for a clean and a polluted example. These are hand-picked individual cases shown for qualitative illustration only; the overall 43.10% validation accuracy and the near-chance confusion matrix remain the authoritative performance measures and indicate that such confident correct outputs are not representative of the model’s general behaviour.

743 4.7. Comparison with Published Studies

744 Compared with published water-imaging studies, the present results are
 745 far lower, but the comparison is not like-for-like. Turbidity-classification
 746 studies report accuracies of 96–99% on large, controlled, validly labelled
 747 datasets [13, 14, 21]. The present study uses a small, in-the-wild dataset
 748 whose labels were found to be randomly assigned, so direct numerical com-
 749 parison is not meaningful. The useful comparison is qualitative: the litera-
 750 ture shows what is achievable with valid data, while this study shows what
 751 happens when label validity fails. Direct comparison is explicitly limited by
 752 differences in dataset, labelling, image splits, and evaluation procedures.

753 4.8. Summary of Research-Question Findings

754 Table 6 summarises the evidence and principal result for each research
755 question. The table shows that the central questions are answered, though
756 several answers are negative or conditional on data quality. No discussion or
757 recommendation is introduced here, in keeping with the template.

Table 6: Summary of research-question findings, listing for each question the principal evidence used, the main result obtained, and whether the question was answered in this phase of the study.

RQ	Evidence	Principal Result	Answered
RQ1	Validation accuracy, confusion matrix	No; near-chance, traced to random labels	Yes
RQ2	Model comparison table	MobileNetV2 marginally higher (50.0 vs 43.1)	Yes
RQ3	Curves, confusion matrix, class metrics	Custom CNN overfits; no class discriminated	Yes
RQ4	Explainability plan	Grad-CAM/SHAP deferred; uninterpretable on random labels	Partly
RQ5	Accuracy, audit, efficiency	Data quality is the binding limit, not architecture	Yes

758 5. Discussion

759 The Discussion interprets the results, connects them to the literature,
760 and assesses strengths, weaknesses, and practical significance.

761 5.1. Interpretation of Dataset and Preprocessing Findings

762 The dataset audit is the most important finding of the study and frames
763 the interpretation of everything else, addressing RQ1. Because the clean and
764 polluted folders were created by randomly splitting a single unlabelled image
765 set, the labels carry no real information about pollution status. This explains,
766 more than any modelling factor, why both models sit at or below the binary
767 chance level. Augmentation and normalisation were applied correctly, but no
768 preprocessing can recover a signal that the labels do not contain. Compared
769 with the controlled-dataset studies in the literature, the difference is not
770 the model but the validity of the ground truth [14, 21]. The finding also
771 highlights how easily a plausible-looking pipeline can hide a fatal labelling

772 flaw, which is a known risk in small projects [15]. The practical lesson is
773 that label validation should precede modelling, not follow it. This reframes
774 an apparent modelling failure as a clear and instructive data-quality result.

775 5.2. Comparative Performance of CNN Models

776 The custom CNN reached high training accuracy but poor validation
777 accuracy, the textbook signature of overfitting, which addresses RQ2. With
778 only 234 images and randomly assigned labels, the model memorised incidental
779 features of individual photos rather than any pollution cue. MobileNetV2,
780 despite a frozen backbone, generalised marginally better, ending at the 50%
781 chance level. This small edge is consistent with the general finding that Im-
782 ageNet features transfer usefully even to narrow tasks [10, 9]. However, the
783 highest validation accuracy here is exactly chance, so it should not be read as
784 evidence of genuine discrimination. The comparison illustrates why the high-
785 est number does not automatically identify the best model, especially when
786 both are at chance. Fine-tuning MobileNetV2 was deliberately avoided to
787 prevent overfitting, so its reported result understates its potential on valid
788 data. The honest conclusion is that neither model learned the intended task,
789 for reasons that lie in the data rather than the architectures.

790 5.3. Class-Level Reliability and Error Patterns

791 The confusion matrix and per-class metrics show errors spread roughly
792 evenly across both classes, which addresses RQ3. No class is reliably recog-
793 nised, and per-class F1-scores cluster near 0.44–0.49, close to what random
794 guessing would produce. There is no strong bias toward either class, which is
795 consistent with labels that contain no learnable structure. Visual similarity
796 between some images, such as reflections or ripples resembling debris, could
797 add confusion, but it is secondary to the labelling flaw. The high-risk conse-
798 quence for the application domain would be a missed pollution event, which
799 an at-chance model cannot avoid. These error patterns reinforce that the
800 binding problem is label validity rather than a subtle architectural weakness.

801 5.4. Interpretation of Grad-CAM and SHAP

802 Explainability is discussed here at the level of what it would and would
803 not show, which addresses RQ4. Had Grad-CAM and SHAP been run on
804 the current models, they would most likely have highlighted regions unrelated
805 to pollution, because the labels are random. That outcome would itself be
806 diagnostic, confirming the absence of a learnable signal rather than revealing

807 meaningful attention. This is precisely why the explainability figures were
808 not produced for this phase: they would be uninterpretable and could mislead
809 a reader. The literature rightly warns that attractive heatmaps are not proof
810 of correct reasoning [11, 12]. Deferring explainability until the labels are
811 valid is the responsible choice and preserves the integrity of the report. Once
812 a validly labelled dataset is in place, Grad-CAM and SHAP will become
813 genuinely informative diagnostics.

814 5.5. *Efficiency and Practical Deployment*

815 Both models are efficient enough to train and run on modest hardware,
816 which addresses RQ5 on the efficiency side. MobileNetV2’s small size makes
817 it attractive for eventual edge or mobile deployment [10]. Efficiency, however,
818 is irrelevant while accuracy remains at chance, so deployment is not justified.
819 A lightweight model is only practical once it demonstrates real discriminative
820 performance on valid data. Any future deployment would require human
821 verification of flagged cases and clear communication of uncertainty. The
822 operational risk of deploying an at-chance model is that it would generate
823 misleading alerts or false reassurance.

824 5.6. *Comparison with Existing Literature*

825 The results neither support nor contradict the high accuracies in the
826 literature, because the experimental conditions differ fundamentally. Pub-
827 lished studies used valid labels and often larger, controlled datasets, achieving
828 strong results [13, 14]. This study’s contribution is methodological rather
829 than performance-based: it documents a labelling failure mode that those
830 studies avoided. Any apparent gap in accuracy is explained by data validity,
831 not by a weakness in the modelling approach itself. Claiming superiority
832 or inferiority on raw accuracy would be inappropriate given the mismatched
833 protocols. The methodological contribution is a transparent template for
834 auditing and reporting small-data environmental experiments.

835 5.7. *Practical and Scientific Contributions*

836 The contributions fall into two categories. On the scientific side, the
837 study provides a controlled custom-versus-pretrained comparison, an explicit
838 data-pipeline audit, and an honest account of a near-chance outcome that
839 confirms a known data-leakage and labelling risk [15]. It also offers a repro-
840 ducible methodology that others can follow and correct. On the practical
841 side, the study demonstrates a low-cost Colab-based pipeline, highlights the

842 importance of label validation before modelling, and shows how to scope ex-
843 plainability honestly. Together these contributions are more valuable than
844 an inflated accuracy claim would have been, because they help practitioners
845 avoid the same pitfall.

846 5.8. *Ethical, Social, and Safety Implications*

847 The principal ethical implication is the duty not to overstate an envi-
848 ronmental tool’s reliability. An at-chance pollution detector presented as
849 functional could divert scarce testing resources or create false reassurance.
850 Because the dataset contains no personal data, privacy and surveillance con-
851 cerns are minimal here. There is, however, a clear obligation to label the
852 system’s status accurately so that no downstream user mistakes it for a vali-
853 dated detector. High-risk environmental decisions must always retain human
854 verification. Reporting the failure honestly is itself the ethically correct ac-
855 tion.

856 5.9. *Limitations*

857 The most significant limitation is the label-construction flaw, which makes
858 the nominal classes meaningless and caps performance at chance. A closely
859 related limitation is the very small dataset of fewer than 300 images, which
860 would constrain learning even with valid labels. The labels were not verified
861 against expert annotation, and no independent test set was held out. Only
862 five training epochs were run, and no systematic hyperparameter search or
863 repeated runs were performed. MobileNetV2 was used with a fully frozen
864 backbone, so no fine-tuning was attempted. Grad-CAM, SHAP, and robust-
865 ness testing were not executed in this phase. These limitations are stated
866 plainly, and future work is described separately rather than as a limitation.

867 5.10. *Future Research Directions*

868 The clearest next step is to obtain or construct a validly labelled dataset
869 in which clean and polluted images are separated by verified content. With
870 valid labels, the same pipeline could be re-run, and Grad-CAM and SHAP
871 would become meaningful diagnostics. Fine-tuning MobileNetV2’s later lay-
872 ers, and trying deeper backbones such as ResNet50 or EfficientNet, would
873 then be worthwhile [22, 23]. A larger and more diverse dataset collected
874 across lighting conditions, water bodies, and cameras would directly address
875 overfitting. Robustness testing under blur, noise, and illumination changes
876 would assess field readiness. Repeated runs with confidence intervals and

877 an independent test set would strengthen statistical reliability. Longer term,
878 a validated model could support low-cost preliminary screening that flags
879 suspect sites for human-verified laboratory follow-up.

880 **6. Conclusion**

881 This study set out to test whether a lightweight CNN could classify water
882 images as clean or polluted on a small public dataset, and how it compares
883 with a MobileNetV2 transfer-learning baseline. The custom CNN learned its
884 training data well, reaching 79.06% training accuracy, but generalised poorly
885 at 43.10% validation accuracy, a clear case of overfitting. MobileNetV2 with
886 a frozen backbone generalised marginally better at 50.00% validation accu-
887 racy, consistent with the value of transferred ImageNet features. The decisive
888 finding, however, came from auditing the data pipeline, which revealed that
889 the clean and polluted folders were created by randomly splitting unlabelled
890 images, so the labels carry no real signal. This explains the near-chance per-
891 formance far better than any architectural factor and reframes the result as
892 a data-quality lesson. Grad-CAM and SHAP were specified but responsibly
893 deferred, because they would be uninterpretable against random labels, and
894 this honest scoping is itself a contribution. The practical conclusion is that
895 label validity and dataset size, not model choice, are the binding constraints
896 for this kind of image-only pollution screening. With a validly labelled, larger
897 dataset and a properly fine-tuned transfer-learning model, the same approach
898 could plausibly perform far better, in line with results reported on controlled
899 water-imaging tasks.

900 **Declarations**

901 The author used a generative AI tool to assist with language clarity and
902 document structuring. The author reviewed and edited all content, verified
903 all reported numbers against the original experimental notebook, and takes
904 full responsibility for the final version of the manuscript. All reported metrics
905 correspond to the actual experimental outputs, and experiments that were
906 not run are reported as such.

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